Beamline Review Report – November 7, 2019

CAT/XSD Group Name: LS-CAT
CAT Director: Alfonso Mondragon/CAT Manager: Keith Brister

Description of the beamline capabilities: LS-CAT administers three beamlines/endstations, 21-ID-D, 21-ID-F and 21-ID-G, equipped with Eiger 9M, and RAYONIX detectors, respectively, which are dedicated to macromolecular crystallography, with both on-site and remote user access. The 21-ID-D beamline is tunable over the range of 6.5-20 keV, while the 21-ID-F and 21-ID-G beamlines are fixed at the selenium edge, with approximate achievable focal beam sizes (FWHM in microns, H×V) of ~10 x 10, and ~50 x 50 and 50 x 50, respectively.

Review committee:
Chair: Steve Almo (Albert Einstein College of Medicine)
SAC Members: Mark Sutton (consultant) & Donald Rej (LANL)
Reviewers:
Martin Fuchs (NSLS-II)
Paul Adams (LBNL)
Phoebe Rice (U. of Chicago)

Overview: The Life Sciences Cooperative Access Team (**LS-CAT**) operates APS Sector 21, and has historically administered three state-of-the-art macromolecular crystallography (MX) beamlines/endstations. LS-CAT operates as a Cooperative Access Team formed by a consortium of eight major academic institutions: Northwestern University, University of Michigan, Michigan State University, Vanderbilt University, University of Wisconsin, University of Illinois (Urbana Champaign and Chicago), Van Andel Research Institute and Wayne State University). Reflective of the substantial and diverse user based, over the past 5 years period 166 PIs have submitted ESAFs, resulting in >2100 PDBs and nearly 750 publications. These resources have had significant impact on structural biology, in general, and, in particular, have greatly enhanced our understanding of fundamental biological processes, including gene regulation, mammalian immunity, cancer biology, RNA biology, viral and bacterial pathogenesis, neurodegenerative disorders, microbiome function and drug design. Based on these contributions, LS-CAT is a premiere beamline in terms of productivity and scientific impact.

To ensure continued, and enhanced operational performance, as well as readiness for future ring upgrades, the management team is carefully evaluating the existing challenges and future opportunities. A strategic plan in now being formulated, including the enunciation of a clear succession plan for long-term management and leadership, which provided confidence that LS-CAT will further expand their reach within the biological community, and establish creative and productive approaches to managing the impending upgrade/"dark period" scheduled to begin Jun 2022. Of particular concern for LS-CAT (and the APS MX community in general) is the establishment of mechanisms for servicing of the current user community, and of equal importance the ability to maintain the exceptional and dedicated administrative and technical staff during this period. LS-CAT leadership is well aware of these challenges, and it is important to highlight the management style that has resulted in a thoroughly engaged and committed staff. This level of staff satisfaction is likely to help in maintaining the team through these uncertain times. Equally challenging for the LS-CAT leadership is the need to maintain the institutional consortium in the face of the upgrade/shutdown and the newly mandated DOE strictures (i.e., 50% GU access). The review panel was highly impressed by the careful, measured and thorough approach taken by the LS-CAT leadership to meet these challenges and uncertainties head-on.

The multiple paths described for sustainability leaves the review panel confident that LS-CAT will continue to service future generation of investigators in the Life Sciences.

Criterion 1: Management Structure **Rating: 1.5**

The management structure is effective, even after recent user staff reductions. Keith Brister is effective in organizing the CAT and the support team. Support staff currently consists of four highly-experienced team members supporting the three beamlines. We sensed good camaraderie within a friendly atmosphere; team members are always available to replace one another when needed, including night support. One staff member commented that the team is the "best group over 20+ years and couldn't imagine a better job." Staff also noted that "Keith is on their side." LS-CAT users highly value interactions with the support staff, and emphatically commented that they have enjoyed nothing but good experiences, including both remote and overnight visits.

Effectiveness of LS-CAT time is high and compares very favorably to local resources:

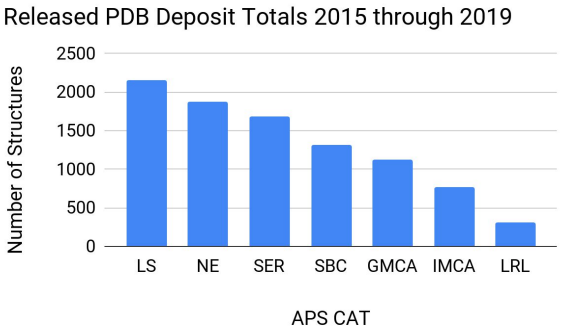


Figure 4: Released PDBs since last review

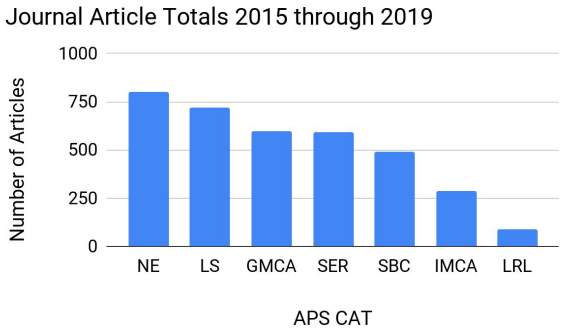
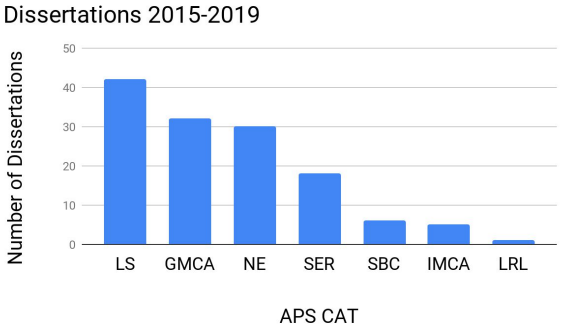


Figure 5: Journal articles since last review



The review panel is concerned that the APS-U project, beginning in 2022, could result in additional reductions to the already "stretched" staff. Management is well aware of this challenge and is working to ameliorate these difficulties; while a definitive plan has not yet been formulated, a number of potential scenarios are under consideration (e.g., support by the participating CAT institutions, participation in local APS-U construction projects and "sabbaticals at other national/international facilities). To further develop this plan, the review panel supports the continued elaboration of a strategic planning document that can be shared with staff, university

administrators, APS management, DOE, NIH, and other sponsors. Such a project management document should also include prioritization of equipment recapitalization, scenarios for cost and schedule estimates, risk management, and succession planning for the LS-CAT leadership. This last point is of particular importance, as Brister will be stepping down as LS-CAT Manager in ~23 months, prior to the anticipated dark period. The entire management team presented a well-considered time-line and set of factors relevant to the recruitment of the new Manager.

Criterion 2: Beamline scientific productivity Rating: 1.0

LS-CAT is highly productive, with the largest number of PDB deposits from APS since 2015 (i.e., over 2000 depositions), the largest number of PhD theses (over 40) and the second-largest number of publications. The general user program is very efficiently run, with a clearly enunciated rapid-access system that is well matched to the research needs of academic laboratories. Although formally 19% of the usage was general users, these may have been undercounted because of time sharing among labs, as this allocation accounted for well over 19% of the PDB deposits and publications. The contributions of the staff and the rapid-access mode to the LS-CAT resources were repeatedly lauded by the users participating in the review. For some groups, in which the PI is not an experienced protein crystallographer, the contributions of the staff are absolutely critical.

In addition to hands-on training for the many students who use their facility, the LS-CAT staff also provide more formal education. For example, they run annual 2-day workshops on data collection and analysis, give tours to high school and undergraduate students, and staff members visit educational institutions to bring the data collection and analysis capabilities directly to the users.

The 9 CAT members make highly efficient use of the facility, with a flexible yet reliable scheduling system that suits their needs and the realities of modern macromolecular crystallography. The review panel feels that LS-CAT is doing a wonderful job of providing regular access to optimized, smoothly run and reliable MX data collection, while also supplying very helpful staff who advise on strategies and help train students. LS-CAT plays a central role in the stellar small molecule screening and drug development program at the University of Michigan, as well as similar efforts elsewhere.

The list provided of the mostly highly cited papers from the review period included several very impactful reports describing the use of structural biology to design and/or understand pharmaceuticals against important medical targets (for example, engineering of broadly neutralizing anti-HIV antibodies and a selective inhibitor targeting a protein driving the progression of leukemia). It is notable that 17% of LS-CAT usage was for proprietary research, underscoring the substantial contributions of LS-CAT to the national biomedical enterprise.

Scientific advances emanating from LS-CAT include a remarkably broad range of areas in pharmaceutical development and the biological sciences. For example, the Ansari and Min groups (Nature Comm, 2015) made the very surprising discovery that a particular DNA repair protein, critical in humans for repairing UV-induced lesions, recognizes those defects through their altered dynamics, and not, as expected, through their static structure. The Nair and Mitchell groups (PNAS, 2018) described how a remarkable group of enzymes catalyzes an extraordinary range of unusual modifications, such as thioamidation, to ribosomally synthesized and post-translationally modified peptides (RIPPs). Data from LS-CAT were critical for defining the chemical mechanism utilized by these enzymes, and have provided the foundation for understanding other distantly related sequences found in the ever expanding sequence

databases, and for leveraging these catalysts to generate a vast array of novel natural product-inspired drug-like compounds.

Criterion 3: Current capabilities **Rating: 1.5**

LS-CAT relies on two canted undulators servicing three endstations. Endstation 21-ID-D is fed by a shortened undulator A, with a tunable energy range (6-21keV), a focal spots size of 20 to 50 microns, an MD2 goniometer, a CAT robotic sample handler and an Eiger 9M detector. The other canted undulator (30mm period) feeds three endstations (21-ID-E, 21-ID-F, 21-ID-G), one of which (21-ID-E) currently does not receive beam. The two x-ray beams feeding the other two endstation are obtained by sequential horizontally diffracting Laue diamond monochromators, and use Be refracting lenses to focus the beam to a 50 micron spot. All three working endstations have identical setups except 21-ID-D has an Eiger 9M; the other two have Rayonic MX-300 detectors. The ability to run three endstations in parallel on one sector takes exceptional advantage of the capabilities of the APS. The fourth endstation, planned for a future upgrade, has an old detector and MD goniometer, which is being used as a test and prototype setup for the potential upgrades to control electronics.

The beamline management and staff have worked to make all three operational endstations as similar as possible, so as to optimize the ability of the limited staff to maintain a high level of functionality and productivity, while allowing for highly flexible scheduling. The current equipment is well maintained, but aging, and it is important that on-going efforts (e.g., multiple pending S10 proposals) to upgrade the robots, detectors and goniometers be vigorously pursued.

The sector's beamlines are comparable to the best of those existing at other synchrotrons around the world. The management has done an excellent job of optimizing their capabilities to satisfy their clientele, which are mostly university researchers. They also have excellent demonstrated records in training and helping students and others new to MX measurements.

Criterion 4: Adequacy and reliability of funding **Rating: 1.5**

LS-CAT has been successful in obtaining funding to support both operations and technical upgrades since their inception in 2002. They are currently funded by a consortium of academic institutions, with 9 main CAT members, principally located in Illinois, Michigan and Wisconsin. In addition, there are several associate CAT members who act through partnership with one of the regular CAT members. For all of these members, efforts are taken by LS-CAT to ensure their research programs are strong and consistent with the technical capabilities of the resource. The benefit of spreading the source of financial support over several entities provides a degree of resilience to fluctuations in support from any single source. This has served LS-CAT well, although it is noted that a recent reduction in support by one member has precipitated a reduction in staffing levels, with one staff crystallographer leaving. This leaves staffing at the minimum level required to support the current user demand, which is very high and very productive. The current CAT members, and their institutions, should appreciate that their collective ability to support the staff directly translates to the productivity of their respective principal investigators.

Although very highly productive, LS-CAT leadership, and beamline staff, recognize that there are technical upgrades needed to enable the beamlines to best deliver results for their users. For example, faster pixel array detectors are needed to increase throughput and data quality on the fixed wavelength beamlines, and the MD2 goniometers need to be replaced with up-to-date versions. The LS-CAT leadership has been proactive in writing NIH S10 proposals for a new detector and a MD3-UP goniometer. Although not always successful on the first application, it is

likely such proposals will succeed over time and help update the beamline infrastructure. As a fallback, LS-CAT leadership is also willing to seek funding from CAT members to help renew instrumentation - this was how the current EIGER 9M was purchased a few years ago.

Looking forward, one of the biggest challenges facing LS-CAT is sustained future funding during and beyond the APS upgrade. The first challenge is the 12-18 month dark period during which X-rays will be unavailable for user experiments. The second challenge is the increase to 50% general user allocation post upgrade (dictated by DOE/BES). LS-CAT leadership is clearly making good progress in planning for both issues and developing scenarios for future funding. The technical improvements post upgrade provide ample reason for the CAT members to continue investing in the resource, and they should expect a significant increase in capabilities to support their science. LS-CAT leadership will need to clearly communicate the advantages to the current, and potential future, CAT members. However, dramatic improvements in beam brightness will make it possible for researchers to study more challenging systems, conduct large-scale screening of samples, and exploit new techniques such as serial crystallography. These capabilities are likely to be important for institutions to attract and retain high-profile structural biology researchers in the future. To better communicate the great potential of the beamlines post APS upgrade, LS-CAT leadership is encouraged to develop a strategic plan to articulate the new capabilities and their impact on the research community.

In summary, LS-CAT has a long history of attracting funding, which has translated in a very high-quality resource with an impressive scientific output. There is every sign that this scientific and funding success will continue in the future.

Criterion 5: Future directions and plans **Rating: 1.5**

The LS-CAT team have enunciated clear plans for the upgrade of the fully tunable endstaton 21-ID-D and for the two fixed energy endstations 21-ID-F and -G. The plan is to keep the D line fully tunable with an adjustable beam size, but reduce the beam size from 20 to 50 micrometers to a beam size from 1 to 20 micrometers.

For the two fixed energy endstations, the planned beamsize is 10 microns, down from 50 um. The review panel strongly supports these upgrades in beamline performance as they are well matched to the expected developments in macromolecular crystallography, with continued movement to smaller crystals, and 10 microns being a good size to collect high quality data from larger crystals using helical data collection methods.

The upgrade of the experimental stations to the newest versions of the respective Arinax microdiffractometers is a sensible approach, since it makes maximal use of the beamline team's expertise with the current MD2 stations, while obtaining state-of-the-art equipment that is well matched to the expected beam sizes and use cases. The MD3-UP diffractometer's specifications are well suited for the challenging one micrometer beam size. Upgrading the current detectors to Eiger2 16M detectors is necessary to make use of the expected high dose-rates of the upgraded beams.

To achieve the demanding beam performance goals, the team have outlined clear priorities. For the 21-ID-D beamline, this means beginning with a refurbishment of the current monochromator, and planning for side-cooled monochromator crystals and repolishing of the focusing mirrors. For the post APS-U period, the team proposes a 30 mm period undulator to realize an achievable 100-fold increase in brightness. Utilizing lenses for beam size changes is a well-established and highly performant approach. Similarly, for the fixed energy stations, 21-ID-F and

G, the current plans for improving the monochromator cooling arrangement, involving the eventually transition to a windowless frontend, is well considered. The medium term plans to provide adjustable focusing optics and work on new robotic sample changers, in coordination with other MX CATS, addresses the need for throughput and robustness and add further functionality.

The team is highly aware of the current developments in the field, such as the move to smaller crystals, and the emergence of new data collection strategies like serial crystallography, while also being highly dedicated to serving their CAT members with a robust and flexible resource. They have begun discussions with other MX beamlines that already push such data collection strategies and have well measured plans for the roll-out of these methods to their members. The team has well established outreach activities, which will help to disseminate these capabilities and to educate the users on these new opportunities. In summary, the LS-CAT leadership is highly cognizant of the upgrades needed to fully support their user community in the new era of APS-U, and they are, yet again, defining a clear and measured plan for the staged implementation of the desired enhancements.

Overall rating for the beamline **Overall Rating: 1.5**

LS-CAT does an outstanding job of serving the MX user community. They have evolved to become a mature facility that is highly reliable, well-run, sustains high-throughput capabilities and contributes considerably to the national biological and biomedical enterprise. The management is particularly commended for their ability to recruit and retain outstanding and dedicated staff who are committed to the success of the beamline and its users. Most notable is the ability of the management team to identify challenges, and not over-react, but to strategically consider and carefully implement a wide range of potential solutions. It is these last characteristics that provide great confidence in the future success of LS-CAT.

Recommendation

The review committee enthusiastically recommends renewal, with the next review to be conducted in accordance with the standard review cycle.